

# Continual Harness: Online Adaptation for Self-Improving Foundation Agents

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**Abstract.** Coding harnesses such as Claude Code and OpenHands wrap foundation models with tools, memory, and planning, but no equivalent exists for embodied agents' long-horizon partial-observability decision-making. We first report our Gemini Plays Pokémon (GPP) experiments. With iterative human-in-the-loop harness refinement, GPP became the first AI system to complete Pokémon Blue, Yellow Legacy on hard mode, and Crystal without a lost battle. In the hardest stages, the agent itself began iterating on its strategy through long-context memory, surfacing emergent self-improvement signals alongside human-in-the-loop refinement. Continual Harness removes the human fully from this loop: a reset-free self-improving harness for embodied agents that formalizes and automates what we observed. Starting from only a minimal environment interface, the agent alternates between acting and refining its own prompt, sub-agents, skills, and memory, drawing on any past trajectory data. Prompt-optimization methods require episode resets; Continual Harness adapts online within a single run. On Pokémon Red and Emerald across frontier models, Continual Harness starting from scratch substantially reduces button-press cost relative to the minimalist baseline and recovers a majority of the gap to a hand-engineered expert harness, with capability-dependent gains, despite starting from the same raw interface with no curated knowledge, no hand-crafted tools, and no domain scaffolding. We then close the loop with the model itself: an online process-reward co-learning loop, in which an open-source agent's rollouts through the refining harness are relabeled by a frontier teacher and used to update the model, drives sustained in-game milestone progress on Pokémon Red without resetting the environment between training iterations.

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## 1 Introduction

Agentic harnesses, the scaffolding that wraps a foundation model with tools, memory, and planning, are now standard infrastructure for autonomous coding agents. Claude Code [2], OpenHands [21], and OpenClaw [19] let models navigate codebases, run commands, and carry state across long interactions. No equivalent exists for embodied agents.

The PokeAgent Challenge [7] reported that without domain-specific scaffolding, frontier

vision-language models make almost no progress on RPG gameplay. Our Gemini Plays Pokémon (GPP) project shows that a human-supervised refinement loop can solve this scaffolding problem:

1

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## Continual Harness

(1) Human-in-the-loop (2) Self-improving harness (3) Model + harness co-learning  
Harness Harness Harness

update

update update

context context context

Refiner

observe trajectory observe trajectory observe  
trajectory

act act act

Environment Agent Human refiner Environment Agent Refiner Environment Agent

weights

RL trainer

Figure 1. Continual Harness automates the harness refinement performed manually in GPP, and extends to joint training of model weights and harness state. Each panel shares the same topology (environment, agent, harness, refiner); only the identity of the refiner changes. (1) Human-in-the-loop: in our Gemini Plays Pokémon (GPP) experiments, a human reads trajectories and rewrites the harness, producing the first AI system to complete Pokémon Blue, Yellow Legacy (hard mode), and Crystal. (2) Self-improving harness: Continual Harness replaces the human with an automated refiner that operates on trajectory data within a single continuous episode; evaluated on Red

and Emerald

across frontier models. (3) Model + harness co-learning: after warm-up stages, an open-source model’s weights and the harness state update jointly during online play.

across Pokémon Blue, Yellow Legacy, and Crystal, we iteratively refined the harness from a screenshot-and-buttons interface into a multi-agent system, and in later runs we removed the human-authored agents and handed the model meta-tools (`define_agent`, `run_code`, notepad edits, custom tool creation) so it could construct its own sub-agents and reusable scripts during play. Our agents beat Pokémon Blue in May 2025, defeated the Elite Four in Pokémon Yellow Legacy on hard mode in August 2025 and completed Pokémon Crystal in November 2025, making

GPP the first AI system to complete multiple Pokémon RPGs. In the hardest stages of Yellow and Crystal, the model itself began iterating on its own strategy through long-context memory, an early emergent form of continual-harness behavior that we formalize and automate in the rest of the paper.

We introduce Continual Harness, a reset-free framework that automates the manual harness refinement of GPP through online in-context learning over the harness state, and extends to joint training of an open-source model’s weights through the same loop. From a minimal environment interface (frame observations, an ASCII text map of the visible area, and button inputs), the agent alternates between acting in the environment and refining its own system prompt, sub-agents, skill library, and memory using trajectory data collected so far in the episode. Every  $F$  steps, a Refiner reads the recent trajectory for failure signatures and runs four passes over the harness applying CRUD edits to system prompt, sub-agents, skills, and memories. Unlike prompt-optimization methods such as GEPA [1] that run complete episodes and reset between updates, Continual Harness updates mid-episode, so self-improvement continues without restarting.

On Pokémon Red and Emerald across three Gemini 3 variants (Pro, Flash, Flash-Lite), Continual Harness substantially reduces button-press cost relative to the minimalist baseline and recovers a majority of the gap to a hand-engineered expert harness, with no curated knowledge, no hand-crafted tools, and no domain scaffolding. On the Emerald cost-vs-completion

Pareto plane, the harness gain scales with model capability: Continual Harness is strictly Pareto-dominant on Pro, high-variance on Flash, and below the capability floor on Flash-Lite.

We then transfer the refined harness to open-source models using an online co-learning loop that scores rollouts with a process reward model, relabeling low-reward windows via a frontier teacher, and updating the model via soft SFT. The online stage closes the loop between harness

refinement and model training. The refined harness shapes the model’s trajectories, and the

## Continual Harness

model’s gameplay surfaces new failure modes for the next refinement cycle. On Pokémon Red, this loop drives sustained in-game milestone progress in an open-source Gemma-4 model across

training iterations, from both beginning and mid-game checkpoints. Both loops operate on the same trajectory data; together they produce continual model-harness co-learning.

Our contributions are: (i) our GPP project results, the first AI system to complete multiple Pokémon RPGs through harness refinement; (ii) Continual Harness, a reset-free framework that assembles harnesses for embodied agents from a minimal environment interface through online in-context learning; (iii) on Pokémon Red and Emerald across Gemini 3 variants, Continual Harness recovers a majority of the gap to a hand-engineered expert harness, with capability-dependent gains on the cost-vs-completion Pareto plane; and (iv) an online co-learning pipeline that drives sustained in-game milestone progress in open-source models on Pokémon Red, producing continual model-harness co-learning.

## 2 Preliminaries

### 2.1 Embodied Agent Environments

We consider an embodied agent that interacts with its environment through a minimal interface. At each timestep  $t$ , the agent receives a frame observation  $o_t$  (a rendered image of the current environment state) together with a text map  $m_t$  that describes the visible tiles and nearby walkable positions in ASCII form, and selects an action  $a_t$  from a fixed set of button inputs  $A$ . The text map is derived from game state that a human player can read off the screen, and compensates for the limited spatial reasoning of current vision-language models; it contains no walkthrough, no objective list, and no pathfinding. The environment is partially observable since the agent cannot access internal state such as NPC intent or battle mechanics beyond what the frame and map expose.

### 2.2 Agentic Harnesses

An agentic harness  $H$  is the scaffolding layer between a foundation model  $M$  and the environment.

Following the decomposition from Karten et al. [7], a harness mediates agent behavior through four components:

- System prompt  $p$ : the instructions and strategic guidance provided to the model at each reasoning step.
- Sub-agents  $G$ : specialized modules that can be invoked by the orchestrator for specific tasks (e.g., battle strategy, puzzle solving, self-reflection).

- Skills K: reusable routines available to the model, spanning both text-level behaviors (heuristics cited in reasoning) and executable programs (pathfinders, tool wrappers). Pre-built primitives such as `press_buttons` and `get_game_state` are skills the harness ships with; new skills can also be authored during play.
- Memory M: a persistent knowledge store that accumulates facts, strategies, and observations across the agent's trajectory.

In addition to these refined components, the harness exposes a fixed set of meta-tools (`define_agent`, `run_code`, `process_memory`, and similar primitives) through which the agent edits  $p, G, K, M$  in place.

A minimalist harness  $H_{min}$  provides only the environment interface ( $ot, mt, at \in A$ ) with a generic system prompt and no sub-agents, memory, or authored skills. A hand-engineered harness  $H_{expert}$  populates all components through manual engineering. A meta-harness gives the model meta-tools (`define_agent`, `run_code`, etc.) to construct its own sub-agents, skills, and memory entries during play; this was the operating point of our later GPP runs, where the

3

## Continual Harness

a Harness refinement within one episode b Co-learning across iterations  
inner loop outer loop

Agent reset-free  
model M

reads  $(st, H_t, \pi_t) \xrightarrow{at} ENV \dots$   
 $at \sim M(\cdot | st, H_t, \pi_t)$

observation  $st$   
harness state  $H_t = (p, G, K, M)$

reads  $iter_k \quad iter_{k+1} \quad iter_{k+2}$

$p$  prompt  $G$  sub-agents  $K$  skills  $M$  memory  
policy  $\{g_i\} \{k_j\}$  tree rollout rollout rollout

$\pi_\theta \circ H_k \quad t \quad \pi_\theta \circ H$

+1 t  $\pi$

k  $\theta \circ H$

k+2 t

Goal: leave Combat move to /red/brock:

Pallet, deliver Handler coords(x,y) “Onix L12; K=256 steps K=256 steps K=256 steps

Oak’s parcel (A, FIGHT) BFS path() use Bulb.”

$\theta_{k+1} = \theta_k - \eta \nabla LSFT$

$\tau, R(st, at, \tau)$

$\Delta$

Refiner model M , every F steps after warmup W

PRM low-R shardDk Soft SFT

Refine: read  $\tau t-F:t$  , emit  $\Delta=(\Delta p, \Delta G, \Delta K, \Delta M R$

); Teacher LoRA on  $\theta$

Apply:  $H_{t+1} \leftarrow H_t \oplus$

pairwise relabel

$\Delta$  3 ep,  $5 \times 10^{-6}$

Figure 2. Methodology overview. (a) Harness refinement within one episode: the Agent reads (st,H,  $\tau$ ) and emits at; every F steps the Refiner reads  $\tau t-F:t$ , emits per-component edits  $\Delta = (\Delta p, \Delta G, \Delta K, \Delta M)$  via the meta-tool API, and  $H \leftarrow H \oplus \Delta$ . (b) Co-learning across DAgger+PRM iterations: each iteration runs  $\pi\theta$  inside a live-refining  $H_t$  for K=256 steps. The trajectory is scored by

k

a pairwise PRM, low-R windows are relabeled by Gemini-3.1-pro, and a soft SFT update produces  $\theta_{k+1}$ .

The loop is reset-free: a persistent state at the end of iter k is loaded as the start of iter k+1.

model built its own pathfinders, battle strategists, and reusable scripts without being asked to. Continual Harness starts from a minimal harness and adds an automated Refiner that rewrites p,G,K,M in place from trajectory analysis. We write HCH for the running harness state during a Continual Harness run, evolving with every refinement cycle.

### 3 Methodology

### 3.1 Overview and Two-Loop Architecture

Continual Harness performs online in-context learning over the harness state  $H$  from Section 2. An LLM Refiner edits  $H$  from the most recent trajectory window during a single continuous episode, generalizing prompt-optimization methods that rewrite only  $p$  from complete-episode resets [1, 14] to a method that rewrites the full state from the trajectory so far.

Write  $st = (ot, mt)$  for the agent’s observation at step  $t$ . The inner loop is the standard agent step: the model  $M$  wrapped by the current harness  $H_t$  produces an action  $at$  from  $st$  and the trajectory so far. The outer loop is harness refinement: every  $F$  steps after a warm-up of  $W$  steps,

a Refiner reads the recent trajectory window for failure signatures and emits per-component edits  $\Delta = (\Delta p, \Delta G, \Delta K, \Delta M)$ . The agent does not reset; the updated harness  $H_{t+1} = H_t \oplus \Delta$  enters the agent’s context on the next step (Figure 2a), with  $p$  replaced by  $\Delta p$  and  $G, K, M$  receiving CRUD-style operations (create, read, update, delete). The Agent and Refiner roles share the same model  $M$ , ablated across Gemini 3.1 Pro, Flash, and Flash-Lite (Section 4). In our GPP runs, the Refiner role for the system prompt and pre-built primitives was performed manually by humans observing the livestream; Continual Harness automates it. Both the agent and the Refiner issue edits through the same meta-tool API (Section 2); they differ only in when each is invoked and on what trajectory context.

### 3.2 Refinement Loop

The Refiner reads  $\tau_{t-F:t}$  and identifies failure signatures over the window: navigation loops, tool-call failures, stalled objectives, and missed exploration opportunities. It then runs four passes, one per component: (i) it rewrites the prompt  $p$  conditioned on the identified failures

4

## Continual Harness

and the trajectory window; (ii) it creates sub-agent entries for repeated multi-step patterns, edits existing entries to address detected failures, and deletes entries that have not been invoked

productively; (iii) it codifies skills from successful sequences and repairs executable code that raised exceptions; (iv) it adds memory entries to fill gaps, updates stale entries, and demotes importance for areas the agent has moved past.

Refinement information accumulates monotonically over the episode: failure signatures observed earlier in the trajectory remain available to all subsequent refinement passes, so refinement quality compounds with episode length, while reset-based methods restart this accumulation after each update. Continual Harness can also target failure modes that only appear deep in an episode (late-game battles, multi-step puzzles, dialogue chains), which reset-based approaches cannot reach by construction since each iteration resets to the initial state.

Beyond these technical advantages, reset-free is also the practically dominant regime for long-running coding agents, embodied agents, and ops tasks where free environment resets are costly or unavailable.

### 3.3 Continual Model-Harness Co-Learning Loop

Figure 2b instantiates Continual Harness as a training loop for an open-source model. After warm-up stages (Appendix D), each online iteration runs  $\pi_\theta$  inside a live-refining harness  $H$

for

$K=256$  steps. A pairwise process reward model (PRM)  $R(st, at, \tau) \in [0, 1]$  scores each transition

over a sliding window of recent transitions (component weights in Appendix D); low-reward windows are relabeled by a frontier teacher, and a soft SFT update on the relabeled shard produces  $\theta_{k+1}$ . The loop is reset-free since the saved emulator state at the end of iteration  $k$  is loaded as the start of iteration  $k+1$ , so the model’s in-game position accumulates across training rather than restarting.

The trajectory distribution  $D_\theta$  depends on  $\theta$  through the harness. The model’s actions induce  $\tau$ , the Refiner reads  $\tau$  to update  $H_t$ , and  $H_t$  in turn shapes the next observation distribution. Both the model weights  $\theta$  and the harness state  $H_t$  are updated by this loop, where  $\theta$  is updated across iterations (via SFT on relabeled trajectories) and  $H_t$  within each iteration (via the Refiner).

## 4 Experiments

We organize our experiments around the contributions from Section 1: our GPP project results (Section 4.2), Continual Harness closing the gap to a hand-engineered harness (Section 4.3), showing improvements with reset-free experience that can bootstrap runs if one does choose to reset, and continual model-harness co-learning for open-source students (Section 4.5). Section 4.6

attributes these gains to in-loop refinement on each of the harness components, and additional details are in the appendix.

### 4.1 Setup

**Environments and metric.** We evaluate on Pokémon Red and Emerald, two RPGs in the same genre that differ in map layout, mechanics, and difficulty. We use the standardized milestone evaluation from the PokeAgent Challenge [7]. The primary metric is cumulative button presses to milestone.

**Harness conditions.**  $H_{min}$ : frames, local text map, buttons, generic system prompt; no sub-agents, memory, or skills.  $H_{expert}$ : the hand-designed harness of PokeAgent [7] and fixed GPP harness with built sub-agents, A\* pathfinding, type chart, damage calculator, and curated objectives.  $H_{CH}$ : starts from  $H_{min}$  and refines during gameplay via Figure 2; three variants: from

scratch, bootstrap frozen (loads a successful from-scratch run, refinement disabled), bootstrap

updating (same bootstrap, refinement continues).

5

Continual Harness

(a) Skill and sub-agent updates (b) Top-5 most-updated components  
pathfinder

3000 Skills 80

Sub-agents gem\_pathfinder\_v2

60 battle\_strategist\_agent

2000 find\_path

40 boulder\_puzzle\_solver

1000 20

0 0

0 50 100 150 200 0 50 100 150 200

Turn (thousands) Turn (thousands)

Figure 3. Yellow Legacy harness refinement is concentrated and recurrent rather than uniform. (a) Counts of CRUD operations (creation, update, delete) on skill and sub-agent definitions, binned per 2,000 turns. The harness is updated throughout the run rather than converging to a fixed scaffold. (b) Update counts for the five most-updated components over the same horizon. A small subset of navigation and battle components accounts for the majority of updates.

Models and seeds. we use Gemini 3 variants (Pro, Flash, Flash-Lite) across all harness conditions, and for open-source transfer (Section 4.5) we use Gemma-4 (E2B, E4B, 26B MoE, 31B dense). We use at least three seeds across all experiments. We report seed medians with per-seed traces at reduced opacity.

#### 4.2 Gemini Plays Pokémon completes multiple RPGs

Our GPP project ran Gemini models live through Pokémon Blue (May 2025), Yellow Legacy on hard mode (August 2025), and Crystal without a lost end-game battle (November 2025), making GPP the first AI system to complete multiple Pokémon RPGs. Since GPP used a mix of human designed and agent iterated harness, we highlight specific cases where we explored

harness refinement over thousands of hours of gameplay.

Emergent Continual Harness behavior through skills. Our Blue-era GPP harness relied on hand-authored specialists such as Pathfinder Agent and Boulder Puzzle Strategist. From Yellow Legacy onward we replaced these with general skills (`define_agent`, `run_code`, `notepad edits`) and let the model build its own harness. Unprompted behaviors included wrapping an `autopress_buttons` sandbox loophole into a general `press_sequence` primitive, developing named multi-stage battle strategies (“Operation Zombie Phoenix” on Crystal’s final Red fight), and authoring an explicit truth-table representation of the Goldenrod Underground switch puzzle in the notepad.

Quantitative harness growth. Figure 3 reports CRUD operations (creation, update, delete) on skill and sub-agent definitions across our Yellow Legacy run. Updates persist throughout the run rather than converging to a fixed harness, and concentrate on a small subset of navigation and battle components. Figure 4 reports structural metrics of one such component, the `battle_strategist_agent` prompt, across successive revisions during the Elite Four phase. The prompt cycles between growth and simplification, and undergoes a structural rewrite in which per-decision logic is absorbed into a `master_battle_agent` that dispatches to named sub-checks. Across both figures the process is the same: a small set of components is repeatedly updated and periodically rewritten. Quality is established separately by GPP’s completion record. We generalize GPP’s mixed methodology to create Continual Harness, which fully automates this process for all modules. Additional results in Appendix B.

#### 4.3 Continual Harness closes the gap to a hand-engineered harness

Figure 5 plots milestones reached against cumulative button presses for Hmin, the three Continual Harness variants, and Hexpert. On both games, Continual Harness substantially reduces the button-press cost of every monitored milestone relative to Hmin and recovers a majority of

6

Lines changed per  
2000-turn bin

Updates per  
2000-turn bin

Continual Harness

(a) Total nodes c1 (b) Decision gates (c) Depth (d) Max fan-out

20 7.5

20 a1 5

5.0

10

10 b1 d1

0 2.5

01 03 05 07 09 11 13 01 03 05 07 09 11 13 01 03 05 07 09 11 13 01 03 05 07 09 11 13

checkpoint checkpoint checkpoint checkpoint

Figure 4. Decision-making complexity of the Yellow Legacy battle\_strategist\_agent prompt at successive revisions during the Elite Four phase of the run: total nodes, decision gates, graph depth, and max fan-out. See appendix B for details.

Pokémon Red Pokémon Emerald  
12 Thunder Badge Knuckle Badge  
10

8 Cascade Badge  
Stone Badge

6 Boulder Badge  
4  
2  
0

0 5000 10000 15000 20000 0 2000 4000 6000 8000  
Cumulative button presses Cumulative button presses

□min CH, from scratch CH, bootstrap frozen CH, bootstrap updating □expert

Figure 5. Milestones reached vs. cumulative button presses. Red (left): 11-milestone subset sequence through Thunder Badge. Emerald (right): 9-milestone sequence through Knuckle Badge (2nd gym); x-axis capped at 8.5k. Lines stop at each run’s last monitored milestone. Thick lines: seed medians; faint lines: individual seeds.

the Hmin-to-expert efficiency gap, without access to the game decompilation, the milestone

schedule, or any of the hand-built sub-agents that constitute Hexpert. The residual gap to the expert harness concentrates in dialogue-heavy gym interiors and multi-turn battle strategy, components Continual Harness does not yet synthesize reliably; we attribute these to specific refinement targets in Section 4.6. On Red, the bootstrap-updating variant is more efficient than from-scratch at every milestone, indicating that the refinement signal compounds within the episode: a harness refined in a prior run accelerates the next even when the game state itself resets. Thus, automated refinement over harness components recovers a substantial fraction of the efficiency of a hand-engineered harness starting from a minimalist interface.

#### 4.4 Continual Harness gain depends on model capability

Figure 6 compares every Emerald run from every model-harness cell with respect to cost and completion. On Pro, Continual Harness is strictly Pareto-dominant over Hmin: from-scratch HCH reaches 100% of milestones at a \$130 median, against Hmin at 98% for \$215, a ~ 40% cost reduction with no completion loss. The two bootstrap variants on Pro reach 96–100% of milestones at \$110–\$140. On Flash, harness benefit is high variance: bootstrap-updating reaches

80% at \$42, marginally above Hmin at 77% for \$30, while from-scratch and bootstrap-frozen variants have a higher variance. Flash-Lite with Hmin reaches 20% at \$11; every Continual Harness variant on Flash-Lite falls to 3–13% at comparable or higher cost. The harness gains requires a model that will sufficiently utilize the harness components properly.

#### 4.5 Open-source students co-learn with a refining harness

We test whether an open-source model improves its gameplay using the self-refining harness with reset-free training (batch size = 1). The model is first primed by supervised fine-tuning on frontier Continual Harness trajectories and an offline GRPO stage on a per-step process reward; neither warm-up stage produces meaningful milestone advancement on its own (Appendix D), and the live in-game gains we report here begin only at the co-learning stage. Each training iteration is a K=256-step DAgger [8, 15] rollout through the full Continual Harness (memory,

7

Milestone

Continual Harness

100%

80%

60%

40% Model (input/output price) Harness

Flash-Lite (0.10 in, 0.40 out USD/M) □ min

Flash (0.30 in, 2.50 out USD/M) CH, from scratch

20% Pro (1.25 in, 10.00 out USD/M) CH, bootstrap frozen

filled = individual seed CH, bootstrap updating

hollow = cell median

0% pareto frontier

101 102

Run cost (USD, log scale)

Figure 6. Emerald cost–completion Pareto plane. Filled markers: individual 24-hour seeds.

Ringed

markers: per-cell medians. Dashed staircase: cost-monotone Pareto frontier. Y axis: fraction of the

31-milestone Emerald set reached; X axis: Gemini API spend (log scale, cached input at 25%).

27 +3

24

15 +3

12

56 ++54

3 +2

1

0 5 10 15 20 25 30

Training iteration k

training-driven advance untrained baseline

Figure 7. Reset-free DAgger+PRM training drives sustained milestone progress on Pokémon Red. Milestone index reached versus training iteration k for the five advancing runs; the broken y-axis

labels each band's start and end. Filled dots: beginning of game. Open rings: mid-game checkpoint.

Stars: judge-verified advances. +N : net objective gain. Dashed line: untrained Gemma-4 baseline (zero

advance beyond the starting milestone). Teacher model: Gemini-3.1-pro.

skills, sub-agents, and prompt all evolving via Figure 2), followed by a process-reward-model scoring pass, a Gemini-3.1-pro teacher relabel of low-reward windows, and a soft SFT update

on the relabeled shard. The training loop is reset-free: the emulator state at the end of iteration  $k$  is loaded as the start of iteration  $k+1$ , so each curve in Figure 7 is a single agent’s in-game trajectory traversed across its own training, not an aggregate over independent rollouts.

Figure 7 shows that the model’s live in-game position advances across training iterations on every plotted run, both from the beginning of the game and from mid-game checkpoints. Both staircase types share the same qualitative shape, indicating that the training signal that drives the model forward from the start of the game also drives it forward from advanced checkpoints; the training procedure is not specific to the early-game distribution. As a negative control, cross-family Qwen3.5 (27B, 35B) without the supervised warm-up stage produces parseable tool calls but cannot leave the starting area in a live rollout (Appendix D.2), ruling out a rollout-protocol artifact. Together with the cross-checkpoint generalization, these results support the co-learning claim: an open-source model trained on data collected from its own play through a continually refining harness improves its in-game position iteration over iteration, without ever resetting the environment. Per-run identifiers, hyperparameters, and the per-iteration process-reward decomposition are reported in Appendix D.4.

8

Milestone index Milestones reached (% of 31)

Continual Harness

270

240

40% 210

180

150

20% 120

90

60

30

0% 0

0 5 10 15 20 25 0 2000 4000 6000 8000 10000 12000 14000 16000

Elapsed hours in run Cumulative button presses

CH, from scratch CH, bootstrap frozen CH, bootstrap updating Dijkstra oracle ☐ min

Figure 8. Pathfinding skill mechanism. (Left) Path-cost deficit of the top-10% evolved navigation skill set (Gemini 3.1 Pro) relative to the Dijkstra oracle over a 24-hour run on warp-to-warp obstacle-navigation tasks; lower is better, dashed line at 0% marks the oracle. (Right) Cumulative navigation-skill invocations against button presses across the same conditions.

#### 4.6 Skills measurably self-improve toward an oracle

We score refined navigation skills by their path cost relative to a Dijkstra oracle. This gives a direct measure of skill self-improvement, independent of end-task efficiency. Figure 8 reports this measurement on warp-to-warp obstacle-aware navigation between fixed map entry and exit points, where greedy open-field hopping fails. Sub-agents, memory, prompt, and reset-free bootstrap transfer are deferred to Appendices C.1 and C.2.

Hmin never invokes a navigation skill; every Continual Harness condition accumulates

hundreds of invocations over a 24-hour run (Figure 8, right). On from-scratch runs the path-cost deficit falls from a near-half-cost penalty at the start to single digits early on and stays there (Figure 8, left). This improvement is in-loop and reset-free: failures from earlier invocations are diagnosed by the Refiner and the affected skills are repaired before later invocations within the same episode. Bootstrap-updating inherits a refined skill set and matches or outperforms bootstrap-frozen throughout, so continued refinement still adds value on top of an inherited set; bootstrap-frozen’s flat trajectory bounds inheritance without further refinement.

### 5 Related Work

#### 5.1 Agentic Harnesses and Scaffolding

Agentic harnesses for coding [2, 19, 21] and assistant tasks [13] stall on embodied RPGs without domain scaffolding [7]. Concurrent prompt-optimization [1, 10, 14] and reflective self-improvement [12, 17] optimize harness components or reflect on trajectories between episodes; Continual Harness edits the full harness state (p,G,K,M) in place mid-episode from partial trajectory windows, without resets.

#### 5.2 Autonomous Agents in Games

LLM-based game agents either build their own tooling during play [3, 20] or pair the LLM with a hand-designed planner [6]. The PokeAgent Challenge [7] provides the canonical embodied-RPG

benchmark and expert harness. Our Gemini Plays Pokémon (GPP) runs across Blue, Yellow Legacy, and Crystal show that human-supervised harness refinement completes multiple full RPGs; Continual Harness automates this process.

#### 5.3 Reset-Free Training, In-Context Learning, and Process Rewards

Reset-free reinforcement learning [4] addresses environments without resets. In-context reinforcement learning [5, 18] and recursive language model methods [23, 25] perform implicit improvement and structured multi-call reasoning over context; Continual Harness writes struc-

tured edits to the full harness state at depth 1. Process reward models [11, 22], group-relative

9

Path cost deficit  
vs. Dijkstra oracle

Nav skill calls

## Continual Harness

policy gradient [16], and STaR-style self-training [24] provide finer-grained signals than sparse episode reward; our co-learning pipeline warms up via SFT and offline GRPO, then runs an online loop where a frontier teacher relabels low-reward windows of the model’s own rollouts inside a live-refining harness for soft SFT updates.

## 6 Discussion

Continual Harness builds and refines its own scaffolding from a minimal environment interface, without resets, and recovers a majority of the gap to a hand-engineered expert harness on embodied Pokémon play. The same alternation runs at both timescales: at inference the agent acts and the Refiner edits the harness in place; at training the online co-learning loop runs the model inside that same live-refining harness, so the trajectory distribution the model learns from co-adapts with its own policy.

A capability floor exists below which the refinement loop cannot bootstrap: Flash-Lite stalls below 20% on Emerald, and every Continual Harness variant on Flash-Lite underperforms the minimalist baseline. Our co-learning experiments couple a frontier-model teacher to an open-source model; the framework extends to the same model serving both roles, but the open-source models we evaluated (Gemma-4 up to 31B) are not yet capable enough to act as both teacher and trainee.

The co-learning loop is not saturated by our experiments: we report sustained milestone progress over the training horizon we ran but did not establish a convergence point. We restrict attention to reset-free training, where the emulator state at the end of iteration  $k$  is loaded as the start of iteration  $k+1$ ; the same loop applies to traditional batch accumulation with resets, and a head-to-head comparison between the two regimes on the same task remains open.

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11

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12

## Continual Harness

### Appendix Contents

#### A Pokémon Environment 14

#### B Gemini Plays Pokémon: Additional Evidence 14

B.1 Yellow Legacy Battle-Agent Evolution Checkpoints . . . . . 15

B.2 Crystal Battle Advisor Evolution Checkpoints . . . . . 16

B.3 Case Study: The Power Plant Route Loop . . . . . 16

#### C Harness Ablations 21

C.1 Mechanism Attribution . . . . . 22

C.2 Reset-Free Bootstrap Transfer . . . . . 24

#### D Training Setup and Results 25

D.1 Training Hyperparameters . . . . . 25

D.2 Gemma-4 Full Eval Matrix . . . . . 25

|                                                        |    |
|--------------------------------------------------------|----|
| D.3 Training Curves and Reward Decomposition . . . . . | 26 |
| D.4 Reset-Free DAgger+PRM Experiments . . . . .        | 26 |

|                       |    |
|-----------------------|----|
| E LLM Acknowledgments | 28 |
|-----------------------|----|

## Continual Harness

### A Pokémon Environment

We run experiments on three Pokémon titles: Pokémon Red (Game Boy, 1996; we use the re-release compatible with the Game Boy Advance emulator), Pokémon Crystal (Game Boy Color, 2000), and Pokémon Emerald (Game Boy Advance, 2004). All three are single-player, turn-based role-playing games with long-horizon structure: overworld navigation, NPC dialogue, turn-based Pokémon battles, inventory management, and gated objectives (badges, plot milestones).

**Interface.** The emulator exposes a screen buffer (rendered at the native resolution of each game: 160×144 for Red/Crystal, 240×160 for Emerald), which we upscale 2× for the vision-language model, and a discrete button channel with eight inputs: UP, DOWN, LEFT, RIGHT, A, B, START, SELECT. Every observation step advances the emulator by a fixed number of frames (120) so that menu animations, battle text, and walking animations resolve between successive agent decisions.

**Text map.** Because vision-language models have known difficulties with fine-grained spatial reasoning over pixel grids, we provide an ASCII text-map `mt` alongside the frame observation `ot` (Section 2). The text map is derived from emulator memory and describes the visible tile grid around the player: walkable tiles (`.`), walls (`#`), interactable tiles (`?`, e.g., signs and talkable objects), NPCs (`N`), ledges, and the player’s position and facing. The map covers the current on-screen area plus a small margin of tiles just off-screen so the agent can plan one step ahead.

The text map contains no walkthrough, no objective list, and no global map information beyond what is currently visible; it compensates for the VLM’s limited spatial reasoning, not for domain knowledge.

**Milestones and the button-press metric.** We use the milestone sequence from the PokeAgent Challenge [7]: a dense, canonical ordering of completion events that a run reaches roughly monotonically as it progresses through the game. Emerald has 31 canonical milestones

through the completion of the 3rd gym, Red has 18 canonical milestones through the completion of the 3rd gym. The primary cost metric is cumulative button presses, not tool calls: a single `press_buttons` invocation emitting the list `[A, A, DOWN]` counts as three presses. This rewards compression in the action channel independent of how the harness structures its tool calls, and it makes Hmin (one button per step) directly comparable to harnesses that batch multi-step presses into one tool call.

Memory reader. The PokeAgent emulator exposes a memory reader that reads structured game state (party, inventory, party HP, current map ID, dialogue text, battle status) from the emulator RAM. Tools such as the expert harness’s A\* pathfinder, battle type chart, and damage calculator use this reader. For Hmin and HCH, the memory reader is exposed only through the text-map `mt` derivation and a handful of general primitives (e.g., `get_party_hp`); not through pre-wired domain tools.

## B Gemini Plays Pokémon: Additional Evidence

Figures 3 and 4 in the main text show Yellow Legacy harness updates and battle-agent structural complexity. This appendix collects the remaining GPP evidence.

Model comparison on Crystal. On Pokémon Crystal under the same harness, Gemini 3 Pro reached comparable early milestones using roughly half the turns and ~ 60% fewer tokens than Gemini 2.5 Pro. The largest divergence occurred at Olivine Lighthouse: 3 Pro initially treated the pits as dangerous, then stepped into one after exhausting safer hypotheses and cleared the

14

## Continual Harness

Emerald Speedrunning Route  
Human Record Pace

0:57 3:00 4:02 4:28 7:58

18 16

31

15

17:27 12:42 12:25 10:53 8:25

12

5 28

19 8 4

18:57 21:28 23:24 25:50 26:30 3

16 17 18 19 20

1

25

Route 116 Rusturf Tunnel Devon Corp Briney's Cottage Dewford Town

29:05 28:31 1:00:57 59:37 27:24

26 25 24 23 22

Oceanic Slateport City Knuckle Badge Dewford Gym Granite Cave +  
Museum Steven Letter 22

20

31:31 36:02 37:24 40:43 43:15 24

27 28 29 30 31

Route 110 Rival Battle Mauville City Mauville Gym Dynamo Badge

Figure 9. Emerald Speedrunning Route. Milestones from Littleroot Town (1) to acquiring the  
Dynamo Badge (31), with game frames from each waypoint. The geographic overview (right)  
maps

key locations. This series of milestones require substantial exploration and backtracking; agents  
must

navigate branching paths, and manage nonlinear dependencies between objectives. The  
current world

record speedrun completes this segment in 1:00:57 minutes [9].

tower, while 2.5 Pro became trapped in a loop of bad assumptions and spent 16,403 turns  
before

obtaining the Fog Badge. Figure 11 shows the corresponding update and fixation patterns.

Failure modes that motivated automation. GPP also exposed recurring failure modes  
that the human refiner had to repair between runs: assumptions made without verification (the

Goldenrod puzzle ran for days because the agent skipped post-battle NPC dialogue containing the missing hint), brittle tool calls with missing parameters, and limited parallel goal pursuit. These are precisely the failures that Continual Harness's mid-episode refinement targets.

#### B.1 Yellow Legacy Battle-Agent Evolution Checkpoints

Figure 4 in the main text plots four graph metrics across the 14 structural checkpoints we extracted from custom\_agents.json over the Elite Four window. For completeness we include the mermaid-rendered decision graph behind each checkpoint across two figures: Figure 12 shows

four canonical checkpoints, and Figure 13 shows the remaining ten. Every chart uses the same renderer and palette, so structural differences reflect prompt changes, not rendering variance. Node colors encode semantic role: entry , analysis , decision gate , terminal action .

15

Continual Harness

Red Speedrunning Route  
Human Record Pace

1:32 2:45 3:05 7:00 7:37

1 2 3 4 5

10

Starter Route 1 Viridian City Route 2 Viridian Forest  
11

17:53 12:09 11:45 9:55 9:44

10 9 8 7 6 6 9

14

Mt. Moon Route 3 Boulder Badge Pewter Gym Pewter City

23:52 34:42 35:53 36:36 38:11 5

11 12 13 14 15 4 15

18

Cerulean City Cerulean Gym Cascade Badge Route 5 Vermillion City  
3

43:04 42:04 38:28  
16

18 17 16  
2

1  
Thunder Badge Vermillion Gym S.S. Anne

Figure 10. Pokémon Red Speedrunning Route. Milestones from obtaining starter Pokémon (1) to acquiring the Thunder Badge (18), with game frames from each waypoint. The geographic overview (right) maps key locations. This series of milestones require substantial exploration and backtracking; agents must navigate branching paths, and manage nonlinear dependencies between objectives. The current world record speedrun completes this segment in 43:04 minutes; see <https://www.speedrun.com/pkmnredblue>.

Opponent Lifetime attempts Notes  
Lorelei 18 Reached Bruno 15+ times  
Bruno 20 Reached Agatha 12+ times  
Agatha 18 Reached Lance 12+ times  
Lance 19 Reached Champion 4 times  
Champion Pixel 4 Victory on attempt 4

Table 1. Yellow Legacy Elite Four lifetime attempt totals. The retries were accompanied by increasingly structured battle prompts (Figure 4) and persistent written memory, producing a text-encoded decision process across the gauntlet.

## B.2 Crystal Battle Advisor Evolution Checkpoints

Like the Yellow Legacy Elite Four window, the Crystal run required heavy prompt iteration during its Battle Tower attempt. We extracted 10 structural checkpoints from `custom_agents.json` for the `battle_advisor` agent. Figure 14 shows the first six checkpoints (turns 30k–33k); Figure 15 shows the remaining four (turns 33k–36k).

## B.3 Case Study: The Power Plant Route Loop

During the Pokémon Yellow Legacy run, the AI agent encountered a 1,003-turn stagnation loop on Map ID 4 (Route 4, near Cerulean City). Spanning approximately 3.5 hours on August 29, 2025, this incident provides a documented example of failure modes in runtime tool generation, schema mismatch during prolonged stalls, and the limits of the agent’s context horizon for error recovery.

The sequence began when the agent decided its goal was to travel to the Power Plant. To do so, it opened the menu to use the HM “Fly.” After repeatedly overshooting its target in the party menu, the agent utilized its tool-generation capabilities to bypass the mechanical inefficiency:

1. It deleted its existing `get_next_pokemon_press` tool.

16

#### Continual Harness

(a) 2.5 Pro updates (b) 3 Pro updates (c) 2.5 Pro fixation (d) 3 Pro fixation  
Skills Skills find\_path battle\_advisor

find\_path  
800 Sub-agents 800 Sub-agents select\_item

20 select\_move 20 force\_press\_bu..  
find\_reachable.. select\_move  
advanced\_route.. press\_sequence

600 600 15 15

400 400 10 10

200 200 5 5

0 0 0 0  
0 20 40 0 20 40 0 20 40 0 20 40

Turn (thousands) Turn (thousands) Turn (thousands) Turn (thousands)

Figure 11. Crystal head-to-head comparison under the same harness. (Top) per-500-turn updates on skills and sub-agents for Gemini 2.5 Pro (left) and 3 Pro (right). (Bottom) top-5 most-updated components per model.

2. It wrote a new tool called `fly_menu_navigator`, setting its `autopress_buttons` flag to `true`.

3. It added a directive to its persistent memory: "I must use the `fly_menu_navigator` tool as intended and trust its output. The `get_next_pokemon_press` tool was deleted to make space for `fly_menu_navigator` and should not have been used. This also highlights a failure to immediately use a newly defined tool."

Schema Mismatch and Execution Loop. The agent invoked the tool, passing in the current screen text and its target destination ("Power Plant"). However, the generated tool call did not match the meta-harness execution schema. To execute a tool with `autopress_buttons: true`, the agent's JSON output requires the `buttons_to_press` array to be explicitly set to `["tool"]`. Instead, the agent logically deduced that it needed to scroll down the list of cities, populating the array with `["Down"]`. Because the meta-harness ignores the `tools_to_call` array if the "tool" button is not explicitly pressed, the system simply pressed Down on the emulator and returned control.

Failing to detect the schema mismatch, the agent recorded in its internal reasoning that the custom tool was successfully executing and returning "Down" as the optimal path. It repeated this exact payload 842 times.

Internal State vs. Execution Reality. Throughout the 3.5-hour loop, the agent's internal thoughts logs indicated it was actively evaluating its progress, despite the environment remaining largely static.

Turn ~172115: "I have completed the mandatory self-assessment... 11. Untested Assumptions & Confirmation Bias: My assumption about the Power Plant being a Fly destination will be tested by cycling through the menu. I am mindful of confirmation bias. My plan for this turn is to continue navigating the Fly menu... I will use the `fly_menu_navigator` tool to take another step towards 'POWER PLANT'."

The agent's internal reasoning indicated it was testing an assumption, but it was caught in a schema error and lacked the necessary game knowledge (the Power Plant is not a valid Fly destination in Generation 1).

Updates per  
500-turn bin

Continual Harness

Start turn analysis

Trust explicit type fields; infer  
species only when type fields are  
missing

Verified type chart is the only  
source of truth

Unknown matchups are forced to  
neutral; immunity is one-way

Survival-first gate

Evaluate current active Pokemon  
before switching

Wild battle during navigation?

Yes Start turn analysis

Default to RUN Use battle\_screen\_text as primary  
local context

No

Immediate context directive  
present?

Fast low-risk OHKO available?

Free turn

Free-turn exception bypasses the HP  
None

gate

1. 4x weakness gate

Yes

2. Status gate on active Pokemon

Move scan across all conscious 3. Critical HP gate

party members

4. Opponent stat-change analysis

5. Prefer invulnerability move when

Search for stronger switch-ins low HP

before committing to a move

6. Action analysis on the current

opponent

Dual-type offense and resistance 7. Do not switch to sleeping  
Pokemon

calculations

No

Move or Switch Other directive

Defensive switching analysis over  
Switch

known moves and likely STAB Start turn analysis

9. Defensive switching analysis

Verified type chart only AI prediction and counter-switch

4x weakness, low-HP weakness, late- risk

game coverage, fragile-avenger, and  
Assess likely STAB damage first

party-state vetoes 1. Action analysis  
Move

Assess known coverage and all known  
moves

Check both STAB branches for dual-

Move or Switch type opponents

Weigh switch cost against immediate  
damage

Best remaining safe action Start turn analysis

Move Switch Switch target must remain Parse inputs  
8. Offensive analysis

offensively viable

Generate type map

2. Offensive analysis 3. Strategic switching Super effective, then STAB, then

Switch target must not be fainted  
neutral Build structured battle state

4. Dual-type offense calculation 5. Survival and threat assessment Decision-making hierarchy

Run Switch Move Explicit dual-type offense Use whole-party HP and status to Follow directive  
before normal  
calculation with immunity collapse rank viable targets strategy

1. Defensive check 2. Offensive check 3. Strategic switching 4. Status moves and setup 5.  
Default offense

Action: RUN Action: SWITCH Action: MOVE Return action  
Return action Return action

a1. Turn 138119: b1. Turn 139085: c1. Turn 151441: d1. Turn 156631:  
baseline veto swarm compact rebuild screen-text + prediction master-agent intro

Figure 12. The four Yellow Legacy battle-agent checkpoints marked a1/b1/c1/d1 on the  
complexity

plot in Figure 4. These span the arc from a linear survival-gate chain (a1), through a hard-reset compact rebuild (b1), to a rebuilt-bigger screen-text-grounded program (c1), and finally to a master-agent decomposition (d1) that dispatches to five named sub-checks.

18

Continual Harness

Start turn analysis

Start turn analysis Start turn analysis Start turn analysis

Trust explicit type fields; infer  
species only when type fields are

Use battle\_screen\_text as primary  
missing

Trust explicit type fields; infer Trust explicit type fields; infer local context

Verified type chart is the only species only when type fields are  
source of truth species only when type fields are

missing missing  
Unknown matchups are forced to

neutral; immunity is one-way

Smaller verified type chart  
Survival-first gate Absolute level-gap veto

Immediate context directive  
Evaluate current active Pokemon

before switching Unknown matchups are forced to present?

neutral; immunity is one-way Survival-first gate

Level-gap veto is hoisted near the Evaluate current active Pokemon  
Wild battle during navigation?

top of the stack before switching  
Free turn

Survival-first gate Free-turn exception bypasses the HP  
None

Yes gate

Default to RUN

No Evaluate current active Pokemon  
before switching

Wild battle during navigation?

Fast low-risk OHKO available?  
Forced-choice desperation state?

Yes

Move scan across all conscious Wild battle during navigation?  
party members Yes

Yes

Search for stronger switch-ins  
before committing to a move Default to RUN Last-stand override can break

No

normal HP and sleep rules

Dual-type weakness, resistance, and  
neutral calculations No

1. 4x weakness gate  
Defensive switching analysis over Yes

known moves and likely STAB

Default to RUN 2. Status gate on active Pokemon

No 3. Critical HP gate

Fast low-risk OHKO available?

Potential switch-in has 4x 4. Opponent stat-change analysis

weakness?

5. Prefer invulnerability move when  
low HP

No Fast low-risk OHKO available?

6. Action analysis on the current

Yes opponent

Low HP plus weakness?

Move scan across all conscious 7. Do not switch to sleeping  
Other directive

Yes Pokemon  
party members

No

Late-game coverage and neutral-hit  
No

risk Search for stronger switch-ins  
Move scan across all conscious Move or Switch

party members before committing to a move

Low-HP revenge-killer plan? Search for stronger switch-ins Dual-type weakness, resistance,  
and Switch

before committing to a move neutral calculations  
9. Defensive switching analysis

No

Yes No Dual-type weakness, resistance, and AI prediction and counter-switch

risk  
neutral calculations

Defensive switching analysis over

Yes Party in critical state? No STAB, known moves, and coverage Assess likely STAB damage first

Move

Defensive switching analysis

Assess known coverage and all known

Yes 4x weakness, failed-escape, low-HP

moves

Yes Reduce risk tolerance to zero No 4x weakness, low-HP weakness, weakness, critical-party-state, and

coverage, fragile-avenger, and fragile-avenger vetoes Check both STAB branches for dual-party-state vetoes type opponents

Switch target 10+ levels lower?

Weigh switch cost against immediate

damage

Switch target must remain

Yes 8. Offensive analysis

Best remaining safe action offensively viable

Reject switch-in Reject fragile switch-in Reject fragile-avenger line Reject switch on level disparity No Best remaining safe action

Super effective, then STAB, then

Switch target must not be fainted

neutral

Best remaining safe action

Explicit dual-type offense If no viable switch exists, Follow directive before normal

Run Run

Switch Move Switch Move calculation with immunity collapse override back into move logic strategy

Move

Switch  
Run

Action: RUN Action: SWITCH Action: MOVE Action: RUN Action: SWITCH Action: MOVE  
Action: SWITCH Action: RUN Action: MOVE Return action

Turn 138914: level-disparity Turn 138916: chart-prune Turn 138923: veto Turn 153601:  
last-stand  
veto reset consolidation strategist

Start turn analysis Start turn analysis

Start turn analysis Immediate context directive  
Immediate context directive

present?  
present?

Free turn  
Immediate context directive

No  
present? Free-turn exception bypasses the HP

None  
Reason only about the current gate

opponent  
1. Status gate on active Pokemon

Start turn analysis 1. Critical HP gate  
2. Critical HP gate

No  
2. Prefer invulnerability move when

Verified type chart only Verified type chart low HP 3. Prefer invulnerability move when  
low HP

1. Action analysis 3. Action analysis  
4. Action analysis

Action analysis

2. Do not switch to sleeping 4. Do not switch to sleeping

Yes 5. Do not switch to sleeping  
Pokemon Pokemon Other directive

Pokemon

Move or Switch Move or Switch Yes Move or Switch Move or Switch

Switch Switch Switch

7. Defensive switching analysis  
Move Switch 4. Defensive switching analysis 6. Defensive switching analysis

Move Move Move Assess likely STAB damage first

Offense priority: super effective, Switch only if clearly better than  
Explicit defensive dual-type Switch-cost nuance and defensive

then STAB, then neutral attacking calculation dual-type threat Comprehensive threat  
assessment  
over all known moves

Switch target must be viable and Switch target must remain  
3. Offensive analysis 5. Offensive analysis Switch target must remain

Basic survival and threat not fainted offensively viable 6. Offensive analysis  
offensively viable

Explicit dual-type multiplication  
assessment

Explicit dual-type offense Follow directive before normal Explicit dual-type offense Follow  
directive before normal

5. Survival and coverage awareness Switch target must not be fainted Explicit dual-type offense  
Follow directive before normal

calculation strategy calculation strategy Switch target must not be fainted  
calculation strategy

Return action  
Return action Return action Return action

Turn 138925: minimal Turn 141323: context + Turn 146358: current Turn 147516: free turn + fallback viability opponent + HP coverage

Start turn analysis

Start turn analysis Parse inputs, including active  
Pokemon status

Parse inputs  
Generate type map

Generate type map Build structured battle state

Build structured battle state Recommend lead

Switch-viability precheck  
Recommend lead

Invalid-switch filter

Decision-making hierarchy

Decision-making hierarchy

3. Strategic disruption and high-  
0. Existential threat 1. Mandatory defensive actions 1.5. Exploit recharge turn 2. Offensive  
knockout 3.5. High-risk switch-in 4. Strategic switching 5. Standard offense

risk plays 3. Strategic disruption and  
0. Existential threat 1. Mandatory defensive actions 1.1. Lone-survivor fallback 1.5.  
Opponent-status exploitation 2. Offensive knockout 3.5. High-risk switch-in 4. Strategic  
switching 5. Standard offense

gambling

Return action plus recommended lead Return action plus recommended lead

Turn 159079: hierarchical master Turn 160511: final master

Figure 13. The remaining ten Yellow Legacy battle-agent checkpoints, grouped by natural aspect.

Row 1: long-chain variants around the first complexity spike and the late “last-stand” rewrite.

Row

2: medium-hierarchy checkpoints across the rebuild-and-grow-again window. Row 3: the two wide master-agent variants that extend the decomposition introduced at checkpoint 12.

19

Continual Harness

Start turn analysis

Counter-user override

Start turn analysis

Expanded team profile with move

typing Start turn analysis

Team profile: Bahamut, Flux, Vortex

Threat ledger for Blissey, Battle Tower legality gate

Alakazam, Starmie, Miltank

Threat ledger and coverage Desperation fallback for near-loss assumptions HP check states

HP check on candidate switch-ins Coverage assumption Gen 2 mechanics block

Status-move sanity check Status-move sanity check Counter-user override

Survival rule for dying active Survival rule for dying active

Compact team profile and role notes

Pokemon Pokemon

Decision FIGHT or SWITCH Decision FIGHT or SWITCH

Decision FIGHT or SWITCH

FIGHT SWITCH FIGHT SWITCH

FIGHT SWITCH

Use Special move only against Switch immediately if only unsafe Use legal attack guided by

Switch to legal teammate with  
Attack with matchup-aware move Switch to best listed resistance Counter users Physical  
attacks remain mechanics and matchup relevant resistance or role

Return action Return action Return action

Turn 30242: baseline matchup Turn 30694: counter mechanic Turn 31220: battle tower legality  
recommender hardening

Start turn analysis

Battle Tower legality gate plus  
held-item awareness

Desperation fallback Start turn analysis  
Start turn analysis

Gen 2 mechanics block Survival check on active Pokemon  
Battle Tower legality gate

Counter-user override Battle Tower legality gate  
Role-based team policy: Rumpus

lead, Vortex ground pivot, Bahamut  
Threat ledger with Lapras and resist pivot Team profile with held items and

Umbreon specifics matchup roles

Threat ledger for Umbreon, Miltank,  
Fatal weakness check before Threat ledger for Gyarados,

Aerodactyl, Blissey, Pikachu  
switching Starmie, Lapras, Scizor, Miltank

Expanded team profile with four Strategy block for lead and swap Strategy block for lead and  
safe  
members patterns switches

Decision FIGHT or SWITCH Decision FIGHT or SWITCH Decision FIGHT or SWITCH

FIGHT SWITCH FIGHT SWITCH FIGHT SWITCH

Use legal move respecting threat- Reject any switch target with known Attack with current role if

matchup Route into the role-specific pivot Fight only if the active Pokemon is Switch to a living teammate from  
specific warnings 4x incoming weakness is favorable for the observed threat alive party\_info

Return action Return action Return action

Turn 31906: fatal weakness check Turn 32755: role-based team policy Turn 33306: survival check

Figure 14. Crystal battle\_advisor checkpoints 1–6 from the Battle Tower window. The graphs trace the early evolution from a baseline matchup recommender (1) through legality and weakness-check additions (2–4) to role-based and survival rules (5–6).

20

Continual Harness

Start turn analysis Start turn analysis Start turn analysis

Start turn analysis

Empirical evidence override Empirical evidence override Empirical evidence override

Opponent-stats prior

Paranoia mode: speed, items, Paranoia mode Paranoia mode with explicit speed

damage, accuracy calculation

Survival check

Survival check

Survival check Survival check

Switch legality check with

mandatory party parsing Switch legality check

Switch legality check Switch legality check

Battle Tower legality gate Battle Tower legality gate

Battle Tower legality gate Battle Tower legality gate

Team profile: Rumpus, Vortex, Team reconfiguration: Rumpus, Flux,

Bahamut Bahamut Final team profile: Rumpus, Flux,

Team profile: Rumpus, Flux, Vortex  
Bahamut

Threat ledger for Steelix, Threat ledger updated for Arcanine,  
Gyarados, Starmie, Lapras, Scizor Starmie, Hitmonchan Expanded threat ledger including

Threat ledger with named counters  
Alakazam and Azumarill

Conservative strategy block with  
Strategy block for lead, elemental

Conservative strategy block Flux and Bahamut pivots Final conservative strategy block  
pivots, and Thunder Wave

Decision FIGHT or SWITCH Decision FIGHT or SWITCH Decision FIGHT or SWITCH  
Decision FIGHT or SWITCH

FIGHT SWITCH FIGHT SWITCH FIGHT SWITCH  
FIGHT SWITCH

Attack only after observed moves Use the current role if observed Route into Flux or Bahamut  
Attack only after exact speed-gap Switch only to a legal teammate  
Switch only if the target survives

Attack under max-stat and speed- Switch only to parsed, living, and pessimistic assumptions  
are evidence and paranoia still permit according to typed threat and item reasoning and  
threat-specific that remains valid under the final  
the paranoid threat model

disparity assumptions role-valid teammates applied it constraints checks speed and item  
assumptions

Return action Return action Return action Return action

Turn 33619: max stats & Turn 34813: empirical Turn 35466: bahamut Turn 36217: explicit speed  
switch legality override & paranoia reconfiguration calculation

Figure 15. Crystal battle\_advisor checkpoints 7–10. The late-window evolution adds  
switch-legality  
bookkeeping (7), an empirical-override / paranoia layer (8), a team-specific reconfiguration (9),

and an explicit speed calculation rule (10).

Loop Termination. The loop terminated after 1,003 consecutive turns once the agent had manually scrolled through all Kanto cities. When the cursor looped back to “LAVENDER TOWN” (its starting point), the agent re-evaluated its state:

Turn ~172322: “I am in the Fly menu, and the current highlighted location is LAVENDER TOWN. My goal is to reach the Power Plant. Since Lavender Town is the current map, I have cycled through all available Fly destinations, meaning the Power Plant is not a direct Fly destination. I need to exit the Fly menu and walk to Route 10...”

The agent stopped calling the custom tool, backed out of the menus, and manually walked north to Route 10.

Observations.

- Context Horizon Limits: Tool generation primarily occurred within the first 50-200

turns of encountering a new bottleneck. Beyond 500 turns in a stall, the agent ceased tool creation and reverted to repeated execution patterns.

- Schema Fragility: When enforcing new rules via persistent memory, the model remained susceptible to schema mismatch, executing actions that did not align with its intended tool design.

- Feedback Blindness: The assumption that the new tool was functioning correctly caused the agent to ignore standard environmental feedback and anomaly detection mechanisms for an extended period.

C Harness Ablations

This appendix gives the full per-component attribution behind the Hmin-to-HCH progression gap (Section 4.6) and the reset-free bootstrap transfer results.

21

Continual Harness

Create-and-forget funnel

add update run: no re-update within 5 run: re-updated within 5 delete Emerald phase 1

Red phase\_1\_1 Red phase\_1\_2 Red phase\_1\_3 created 99  
nalv\_0ig0a0t1e navigate\_to invoked  $\geq 1$  17 (783 invocations)

move\_to\_coords move\_to\_coords  
gog\_virindas

iav  
k  
nig  
il  
\_aroteu\_tev12

invoked  $\geq 3$  11  
succeeded  $\geq 1$  5 (304 successful)

o\_viridianna\_vm\_art skill\_0002 navigate  
skill\_0001  
skill\_0003 skill\_0002

n n  
naav

avd\_fabsfsc  
nav

—  
v\_

r walk\_to skill\_0004 Red phase 1  
\_r  
i  
poueattthe

tt skill\_0004  
skill\_0005 skill\_0005 created 110

nasvk\_isl \_ 0 3 invoked  $\geq 1$  33 (487 invocations)  
\_t  
l  
sria

0 bfs\_move navigate\_v2  
mig0phlet

nm  
nav

avo\_vneo\_rutp 0 skill\_0008 invoked  $\geq 3$  18  
safe\_navigate pathfinder

ah travel\_to run\_route\_1\_north  
r n v\_

\_  
\_1  
1 succeeded  $\geq 1$  3 (247 successful)  
t0o

so t  
kk  
b s  
ii  
u b  
ll

s  
ll\_\_00

\_ skill\_0012  
0000

f  
84 cave\_bfs smart\_navigate Emerald phase 2 cont.

smart\_cave\_bfs  
cave\_router safe\_path\_executor

ss  
sk\_created 104  
kk  
i 5  
ii  
l 0  
ll  
l  
ll\_\_00

0  
0010

0  
27

ss  
sk \_ smart\_navigate skill\_0003 invoked  $\geq 1\ 16$  (364 invocations)

city\_nav  
kk  
i 5  
ii  
l 0  
ll  
l  
ll\_\_00

0  
0021

1 invoked  $\geq 3\ 11$   
89

ss  
ss

skill\_0015 skill\_0008  
k \_ skill\_0009 skill\_0011 succeeded  $\geq 1\ 5$  (73 successful)

kk  
i 9  
ii  
l 0  
ll  
l  
ll\_\_00

0  
0022

2  
07 skill\_0016

go\_to\_coo  
ss  
s  
mk  
kill

rds skill\_0009  
skill\_0001

all  
l 0 1  
r\_  
\_0  
t0

3 Red phase 2 cont.  
created 335

\_0n1a3v skill\_0017  
skill\_0020  
skill\_0025 invoked  $\geq 1\ 53$  (720 invocations)

ss  
s l  
kk  
k 3  
ii  
i 0  
ll  
l skill\_0020  
ll\_\_  
\_4  
0000

0  
3387 diglett\_nav skill\_0023 invoked  $\geq 3\ 31$

succeeded  $\geq 1\ 14$  (380 successful)  
0 2500 5000 7500 0 2000 4000 0 2000 4000

Step Step Step 0 100 200 300 400  
Skill count

Figure 16. Left: per-seed skill lifetime for the three Red from-scratch HCH runs. Markers are add, update, run\_skill (green if no re-update within 5 steps, red otherwise), and delete. Right: create-and-forget funnel across both games and both HCH bootstrap variants.

## C.1 Mechanism Attribution

### C.1.1 Pathfinding skills

Figure 8 in the main text shows the Dijkstra-oracle ratio and cumulative skill calls. Below we give the measurement details and the residual structure the main-text summary omits.

For each first-traversal segment between consecutive milestones, we compute a BFS-optimal path length on the union of tiles observed by any run of that game on that map, and divide the agent's issued button presses within the segment by that optimum. Dialogue and battle presses are filtered out so the comparator only sees navigation. The ordering inverts inside gyms

where dialogue and puzzle state dominate over navigation, which is why the residual gap to Hexpert in the main-text progression figure concentrates there. The refined library is heavily biased toward BFS and A\* wrappers because saved presses on navigation translate directly into

faster milestones, the strongest local signal the refinement loop sees. Hexpert routes navigation through a pre-built A\* tool not visible to the run\_skill counter, which is why its curve sits at the floor in the main-text right panel. Most of the Hmin-to-HCH delta is absorbed by the skill library alone.

### C.1.2 Skill debugging

For every persisted skill that sees at least one skills\_updated event in the refinement log, we compute the rolling success rate over a window of invocations immediately before and immediately after each update. We also track the create-to-succeed funnel: skills authored, invoked at all, invoked repeatedly, and ever successful. Reset-free refinement produces dramatic

repairs on the skills the agent actually relies on, and the repair happens in the same episode where the failure occurred. Figure 16 shows this with two complementary views. The left panel is per-seed skill lifetime: each skill occupies one lane, markers are add/update/run/delete events,

and the update-to-next-execute edge is drawn so each debug iteration reads as an update followed

by a run. The right panel is the create-and-forget funnel: most authored skills are never invoked, a small working set absorbs the bulk of calls, and even fewer see success. The refinement loop therefore triages: it repairs the skills the agent depends on, tolerates regressions on unused ones,

and accepts a long create-and-forget tail. This is the argument for reset-free operation over

reset-based baselines: the failure record and the repair sit inside the same trajectory, so the loop closes within a run rather than across resets.

22

## Continual Harness

(a) cost by role (b) handoff count (c) handoff success by task  
175 CH, from scratch

CH, bootstrap updating dialogue 100 67  
107 150

navigate 83 65  
125

106

11x (Emerald p1) 100 battle 83 23  
105 8x (Red p1)

75 grind 62 25

104 50  
orchestrator menu 100 100

103 subagent 25  
CH, from scratch catch 65 15  
CH, bootstrap updating 0

0 2500 5000 7500 0 2500 5000 7500 exit focus  
step step

Figure 17. Sub-agent handoffs. (a) Cumulative approximate tokens by role (orchestrator solid, sub-agent dashed). (b) Cumulative execute\_custom\_subagent count. (c) Per-task-type handoff success: exit is the percent of spans ending via return\_to\_orchestrator; focus is the percent of returns where the orchestrator either pursued the pre-handoff objective or crossed a milestone within ten subsequent steps.

### C.1.3 Sub-agent handoffs

We segment each run into spans of orchestrator execution and sub-agent execution, track approximate per-step input tokens from prompt length, and label each sub-agent span by task type. Sub-agent handoffs serve two roles for the harness: they keep per-step cost low by giving the sub-agent a narrow specialized context, and they let the orchestrator resume its prior objective after the sub-agent returns. Figure 17 makes both points visible. The left panel plots orchestrator tokens and sub-agent tokens on a shared step axis; the sub-agent curve sits about an order of magnitude below the orchestrator curve throughout, which is the per-step saving the harness buys by partitioning context. The middle panel tracks cumulative handoff counts per condition; bootstrap-updating tracks from-scratch closely throughout the run, with a small plateau gap by run end. The right panel scores post-return behavior per task type; clean-return and on-task-recovery rates sit near the top of the scale for navigation, dialogue, and menu tasks.

The harness rather than the raw model carries most of the long-horizon performance: once the orchestrator can delegate to cheap specialized contexts and trust the return, long tasks become tractable with far fewer tokens than the raw context would imply.

### C.1.4 Memory reuse

Every orchestrator step prompt lists the IDs and titles of all stored memories under a LONG-TERM

MEMORY OVERVIEW section, so the agent sees the full catalog for free. The question is whether the

agent pulls on that catalog: requests the full content of an entry via `process_memory`, invokes a memorized skill, or cites an entry ID in its reasoning. We measure this pull rate as the fraction of available entries ever referenced in an episode, and we localize reads to the milestone windows

where the agent should need them: Mauville Gym on Emerald, Mt Moon on Red. Memory is leveraged once the library is both mature and inherited (Figure 18): bootstrap runs, which load a from-scratch memory store at the start, consult it actively inside the gym and cave segments, while from-scratch runs write many entries and rarely reach back for them. The reference rate remains low in absolute terms, which we report honestly; most authored entries sit unused. The transferable unit of the framework is therefore the harness across runs, not a single episode, and

23

cumulative tokens (prompt chars / 4)

cumulative handoffs

## Continual Harness

Mauville timeline Mt Moon timeline Mauville ops Mt Moon ops

□ min n=1, reuse 0% n=0, reuse 0%

CH, from scratch n=16, reuse 100% n=72, reuse 35%

CH, bootstrap frozen n=2, reuse 100% n=4, reuse 100%

CH, bootstrap updating n=10, reuse 100% n=20, reuse 100%

0 2000 4000 0 500 1000 1500 0 20 0 50

Steps in window Steps in window Memory ops Memory ops

loaded from bootstrap written earlier this run written in the bottleneck target unresolved  
background refinement  
add update read delete

Figure 18. process\_memory use inside the first real bottleneck of each game: Mauville Gym (Emerald, left column) and Mt Moon (Red, right column). One representative seed per condition. Top row: per-run timeline where each marker is a process\_memory tool call at the step it fired; color encodes provenance of the memory entry. Bottom row: the same ops aggregated into a stacked composition bar.

an explicit reuse prior is a natural next step.

### C.2 Reset-Free Bootstrap Transfer

The bootstrap-updating variant tests whether the transferable unit is a single episode or the harness across episodes. On Emerald, every store the agent exercises during a bootstrap-updating run still targets the inherited harness. On Red, memory and skill invocations stay inherited, but the bootstrap-updating agent collapses its sub-agent budget to a handful of calls, and the few it makes cite IDs not present in the bootstrap. When that collapse happens, the milestone staircase regresses. The harness-as-transferable-unit claim therefore holds when the agent continues to exercise the inherited components and breaks when it abandons them: a reuse prior or a sub-agent deletion policy is the natural next step.

Bootstrap runs load the final skills, sub-agents, and memory of a successful from-scratch run before any new play. For every invocation we classify whether the invoked entry originated in the bootstrap or was authored during the bootstrap run itself, using retrieval semantics that

count actual use across stores rather than passive prompt inclusion. Bootstrap succeeds by leveraging the inherited harness, and regressions show up where the agent stops using it. The end-of-run inherited share per store is reported in Table 2. On Emerald every store that the agent exercises targets the inherited harness, with substantial per-run invocation counts on skills

and sub-agents. On Red, memory and skill shares also stay inherited, and the only anomaly is sub-agents: the Red bootstrap-updating agents collapse their sub-agent budget to a handful of calls and the few calls they make cite IDs not present in the bootstrap. The milestone staircase in Figure 5 reads this phenotypically: Emerald bootstrap tracks from-scratch closely, while Red bootstrap-updating drifts below from-scratch and then below Hmin in lockstep with the collapse of sub-agent use. The harness rather than the episode is the transferable unit:

reset-free

refinement carries across runs when the inherited components stay in play, and it breaks when the agent abandons them.

### C.2.1 Red Bootstrap-Updating Regression

The Red bootstrap-updating regression visible in Figure 5 begins around step 213. Newly authored sub-agents overtake the inherited ones at that point. They have not gone through the repair cycle observed for from-scratch skills (Figure 16), so their per-invocation success rate sits below what the bootstrap sub-agents reached before they were cached. A reuse prior on

24

## Continual Harness

Table 2. C5 inheritance: fraction of phase-2 invocations whose target was present in the phase-1 bootstrap. Mean across seeds (n=3 for each cell). “—” means no invocations of that store type in the run.

### Game Store Frozen Continued

Emerald skills 100.0% ± 0.0 99.6% ± 0.6

subagents 100.0% ± 0.0 100.0% ± 0.0

memories 98.2% ± 1.9 100.0% ± 0.0

Red skills 100.0% ± 0.0 96.5% ± 3.8

subagents 100.0% ± 0.0 6.4% ± 5.7

memories 100.0% ± 0.0 100.0% ± 0.0

sub-agent selection, or a deletion rule that deprecates newly authored sub-agents whose task

signature is covered by inherited ones, are natural follow-ups.

## D Training Setup and Results

This appendix gives the full training details for the open-source transfer pipeline (Section 3.3): SFT, offline GRPO, and online co-learning hyperparameters; the full evaluation matrix across Gemma-4 sizes; and training-curve diagnostics for the co-learning stages.

### D.1 Training Hyperparameters

Supervised fine-tuning. We fine-tune Gemma-4 variants (E2B, E4B, 26B MoE, 31B dense) via LoRA with rank  $r=256$ ,  $\alpha=256$ , bf16 precision, and an 8K-token context using Unsloth on H200 GPUs. Each example is a (screenshot, harness prompt, teacher response) tuple extracted from Gemini-3.1-pro Continual Harness gameplay. Learning rate  $2 \times 10^{-5}$ , linear warmup over 3% of training, cosine decay. We train for one pass over the teacher-trajectory set per model.

Offline GRPO. For each state in the teacher-visited set, the SFT-initialized policy generates  $G=4$  candidate completions. Each is scored independently by a Gemini-3-flash-preview per-step oracle on a composite of action correctness (binary, weight 0.6) and format compliance (binary, weight 0.4). Advantages are group-normalized within the  $G$  samples per state and the policy is updated via standard GRPO [16]. Learning rate  $1 \times 10^{-6}$ , KL coefficient  $\beta=0.04$  against the SFT reference, batch size 8 states per optimizer step, 590 total optimization steps.

Online co-learning loop. Each online iteration is a  $K=256$ -step DAgger [8, 15] rollout through the full Continual Harness (memory, skills, sub-agents, and prompt all evolving via Figure 2) on Pokémon Red. A pairwise process reward model [22] (Gemini-3-flash-preview) scores each transition over a sliding window; reward is a weighted combination of trajectory progress (0.4), action correctness (0.3), reasoning quality (0.2), and format compliance (0.1). Low-reward windows are relabeled by a Gemini-3.1-pro teacher, and a soft SFT update on the relabeled shard produces the next iteration’s checkpoint.

### D.2 Gemma-4 Full Eval Matrix

The main-text claim that neither warm-up stage produces meaningful milestone advancement (Section 4.5) is documented here per Gemma-4 size. The matrix covers E2B, E4B, 26B MoE, and 31B dense. Smaller sizes converge to low SFT training loss but collapse to `tool_format=0` on the real harness prompt, consistent with an interaction between SFT signal strength and the 8K context needed to hold the full state plus reasoning. The 31B SFT Emerald model underperforms the 26B SFT Emerald model on most metrics in our runs.

## Continual Harness

Table 3. Full Gemma-4 eval matrix on Emerald. SFT rows are fine-tuned on Gemini-3.1-pro Continual

Harness trajectories. The GRPO column reports the offline-GRPO warm-up checkpoint, which emits degenerate completions on this prompt set. Smaller Gemma-4 sizes train to low loss but collapse under the full harness prompt.

Metric Base SFT (emerald) GRPO  
26B 31B e4b e2b 26B 31B e4b e2b 26B

|                      |      |      |      |      |      |      |      |      |      |
|----------------------|------|------|------|------|------|------|------|------|------|
| tool_format          | 0.00 | 0.00 | 0.00 | 0.00 | 0.95 | 0.35 | 0.00 | 0.00 | 0.00 |
| actionable           | 0.20 | 0.55 | 0.55 | 0.40 | 0.95 | 0.35 | 0.40 | 0.20 | 0.00 |
| grounding            | 0.15 | 0.39 | 0.54 | 0.28 | 0.72 | 0.45 | 0.52 | 0.33 | 0.40 |
| action_relevance     | 0.15 | 0.51 | 0.38 | 0.15 | 0.50 | 0.25 | 0.35 | 0.07 | 0.00 |
| reasoning_similarity | 0.15 | 0.45 | 0.26 | 0.10 | 0.35 | 0.05 | 0.28 | 0.05 | 0.00 |
| hallucination        | 0.05 | 0.05 | 0.30 | 0.05 | 0.55 | 0.50 | 0.30 | 0.25 | 0.00 |
| degenerate           | 0.00 | 0.00 | 0.00 | 0.05 | 0.05 | 0.10 | 0.10 | 0.00 | 0.00 |
| tokens/s             | 154  | 18   | 188  | 221  | 166  | 27   | 188  | 221  | 26   |

Table 4. Full Gemma-4 eval matrix on Red. The 26B Red SFT row is omitted because the adapter was degenerate at eval time. 31B SFT is the viable Red checkpoint and is used as the initial policy for the online co-learning stage.

Metric Base SFT (red) GRPO  
26B 31B e4b e2b 31B e4b 26B

|                      |      |      |      |      |      |      |      |
|----------------------|------|------|------|------|------|------|------|
| tool_format          | 0.05 | 0.05 | 0.10 | 0.00 | 0.50 | 0.10 | 0.50 |
| actionable           | 0.25 | 0.35 | 0.45 | 0.15 | 0.50 | 0.35 | 0.50 |
| grounding            | 0.23 | 0.31 | 0.40 | 0.09 | 0.44 | 0.38 | 0.44 |
| action_relevance     | 0.33 | 0.42 | 0.33 | 0.03 | 0.75 | 0.55 | 0.65 |
| reasoning_similarity | 0.25 | 0.40 | 0.40 | 0.00 | 0.65 | 0.45 | 0.50 |
| hallucination        | 0.05 | 0.00 | 0.25 | 0.20 | 0.30 | 0.30 | 0.40 |
| degenerate           | 0.00 | 0.00 | 0.00 | 0.05 | 0.00 | 0.00 | 0.05 |
| tokens/s             | 162  | 21   | 189  | 233  | 26   | 101  | 89   |

D.3 Training Curves and Reward Decomposition  
See Figure 19.

#### D.4 Reset-Free DAgger+PRM Experiments

The online co-learning loop in Section 3.3 composes a DAgger-style teacher-relabel step [8, 15] with a pairwise process reward model (PRM) and reset-free emulator-state propagation across iterations. Each iteration’s K-step rollout begins from the saved emulator state of the previous iteration, the PRM scores per-step pairs, the Gemini teacher relabels low-reward windows, and

the model is updated via soft SFT on the relabeled shard before the next rollout. We instantiate this on Pokémon Red with the 26B Gemma-4 SFT model as the initial policy and study whether multiple iterations of training advance the agent through the in-game milestone progression. Each run targets one starting checkpoint along the Red progression, with starting points spanning the early game (milestone 1, leave Pallet Town) through the mid game (milestone 24, defeat rival in Cerulean City; milestone 30, meet Bill).

Run variation. We run a set of training jobs in parallel and vary three design choices. Some runs start from the beginning of the game and others load a saved emulator state at a fixed mid-game milestone. Across the beginning-of-game runs we vary the shard-window size and the shard accumulation policy. One run replaces the default relabel teacher prompt with a variant that injects already-completed objectives into the context. All runs use rollout-step budget  $K=256$ , soft SFT (3 epochs at  $5\times 10^{-6}$ ), and pairwise PRM with stride 8.

Cumulative game progress over training. Figure 7 reports the milestone index reached versus training iteration for the advancing runs. Mid-game starting points advance from

26

### Continual Harness

Table 5. Progressive improvement across warm-up stages on Pokémon Red, evaluated on 20 held-out transitions. SFT lifts format compliance from near zero. Offline GRPO with a 4-component heuristic reward and offline GRPO with a Gemini-oracle reward both maintain format and shift action quality. The online co-learning stage produces sustained milestone progress in live gameplay; per-iteration progression and PRM rewards are reported in Section D.4. Qwen3.5 35B is shown as a cross-family baseline: it produces parseable tool calls through the harness but cannot advance in the game.

| Format (Tier 1)              | Action Quality (Tier 2) |          |         |         |        |        |  |
|------------------------------|-------------------------|----------|---------|---------|--------|--------|--|
| Stage                        | Model                   | tool_fmt | act'ble | act_rel | reason | ground |  |
| Base                         | Gemma-4 26B             | 0.05     | 0.25    | 0.33    | 0.25   | 0.23   |  |
| SFT                          | Gemma-4 31B             | 0.50     | 0.50    | 0.75    | 0.65   | 0.44   |  |
| Offline GRPO (heuristic)     | Gemma-4 26B             | 0.50     | 0.50    | 0.65    | 0.50   | 0.44   |  |
| Offline GRPO (Gemini oracle) | Gemma-4 26B             | 0.50     | 0.50    | 0.55    | 0.30   | 0.40   |  |

Base Qwen3.5 35B parseable 0 game progress (stuck)

(A) SFT lift across Gemma-4 sizes (B) Offline GRPO reward curves

1.0

1.0 Emerald base Red base Offline GRPO Emerald (heuristic)

Emerald SFT Red SFT Offline GRPO Red (heuristic)

0.8 Offline GRPO Red (Gemini oracle)

0.8 SFT baseline

0.6 0.6

0.4 0.4

0.2 0.2

0.0 0.0

26B 31B E4B E2B 0 100 200 300 400 500 600

MoE training step

Figure 19. Gemma-4 tool-calling behavior across warm-up stages. (A) SFT on frontier Continual Harness trajectories lifts tool\_format success from near zero. (B) Reward curves for the two offline

GRPO variants (heuristic 4-component reward and Gemini-oracle reward); the dashed line marks the approximate SFT reward baseline.

their loaded indices, indicating that the training procedure is not tied to the early-game distribution. Advances are bursty, with cumulative milestone gain accumulating across multi-iteration improvement bands rather than within single iterations.

Per-iteration PRM reward. The per-iteration PRM reward is non-monotonic across all runs. Reward sustains of multiple consecutive iterations near or above 0.40 precede the largest milestone gains, with regression iterations interleaved; the bursty advance pattern in Figure 7 aggregates these reward-sustained windows.

Resume-checkpoint regression. The first iteration after a checkpoint resume regresses relative to the pre-resume iteration and recovers within two to three iterations. We treat this as an artifact of the resume protocol; we compute the aggregate signal over resume-spanning windows. Milestone advances occur in post-resume iterations, indicating that the regression is reward-specific and does not propagate to the trajectory-completion signal the judge uses.

27

tool-format success

mean reward (normalized)

Continual Harness

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28